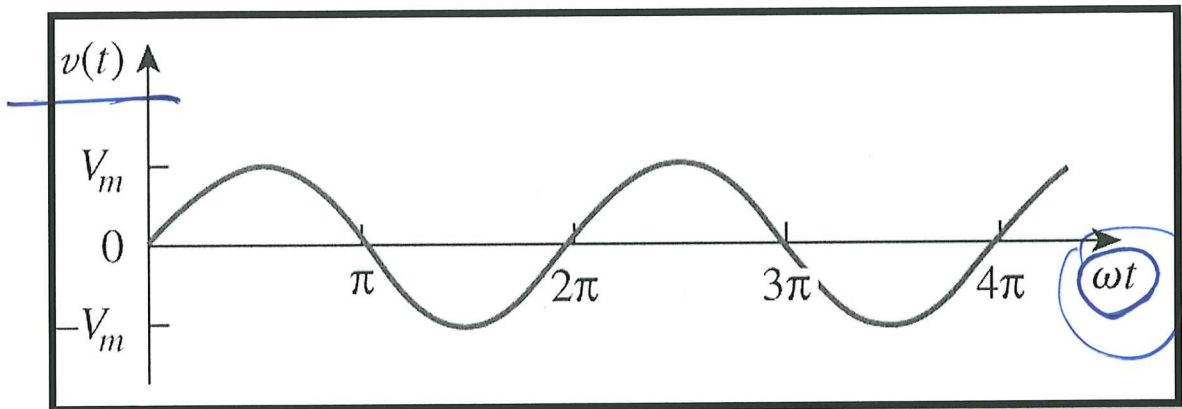


L#25: Sinusoids

A sinusoid is a signal has the sine or cosine functions.



$$v(t) = V_m \sin(\omega t + \phi)$$

V_m : Amplitude of sinusoid

ω : angular frequency in rad/s

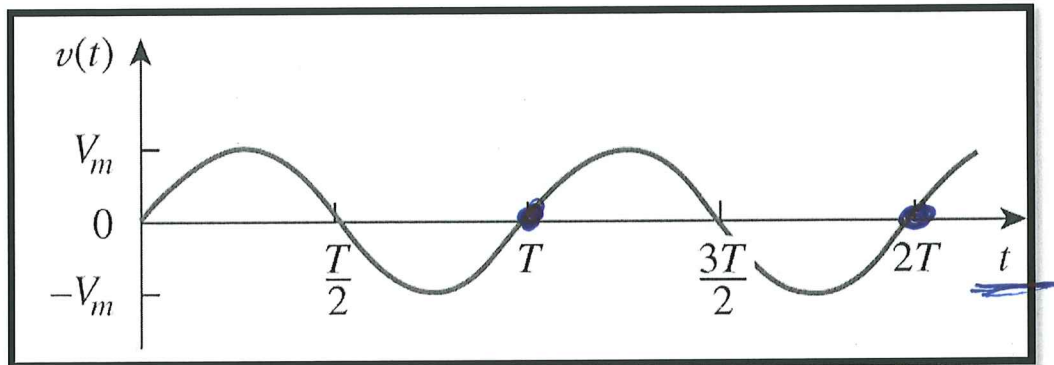
t : time in second [s]

ωt : argument of the sinusoid [rad]

ϕ : phase of sinusoid, rad or degrees

→ radians

Sinusoids vs. time (not ωt)



T is period [s]

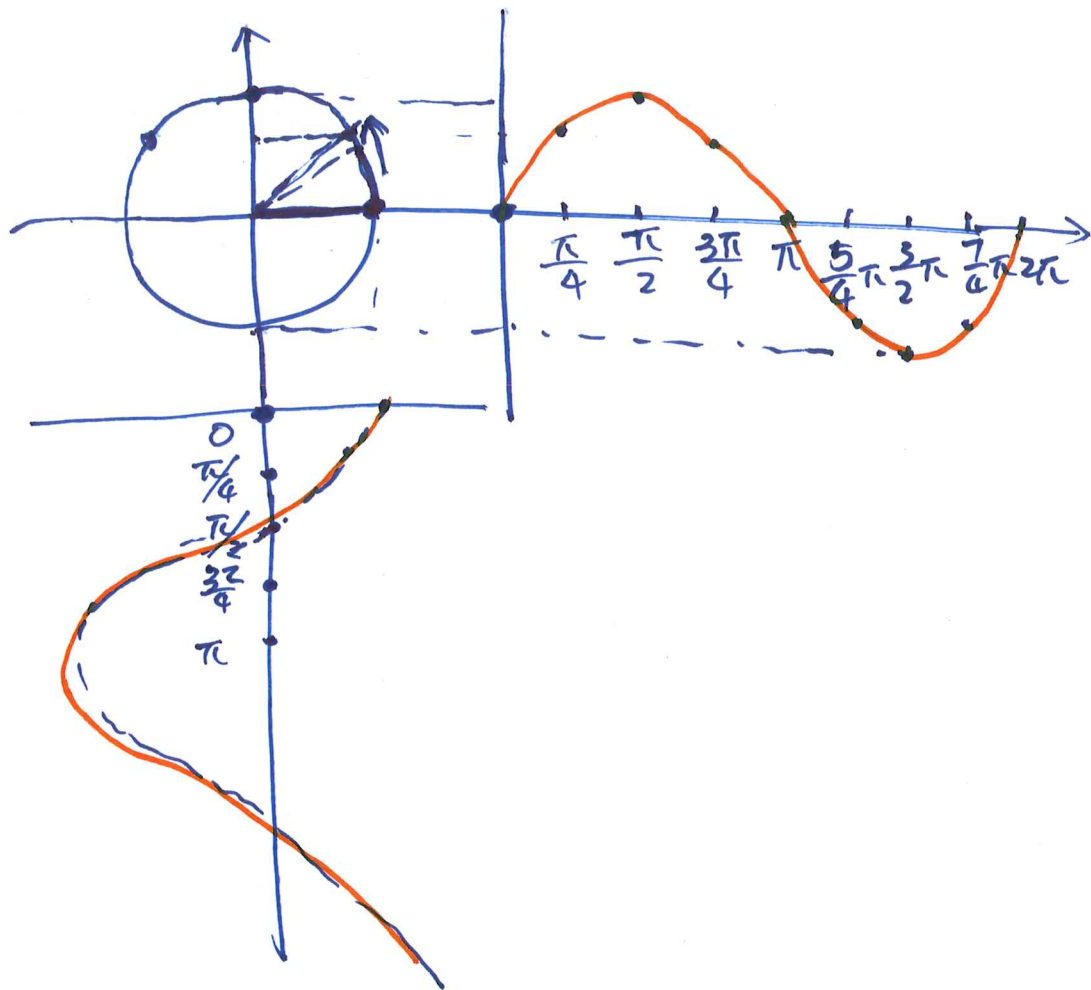
At $t=T$: $\omega t = 2\pi$, $\omega T = 2\pi$
 $\Rightarrow T = \frac{2\pi}{\omega}$ [s]

f: $f = \frac{1}{T} = \frac{1}{2\pi/\omega} \Rightarrow f = \frac{\omega}{2\pi}$ [Hz]

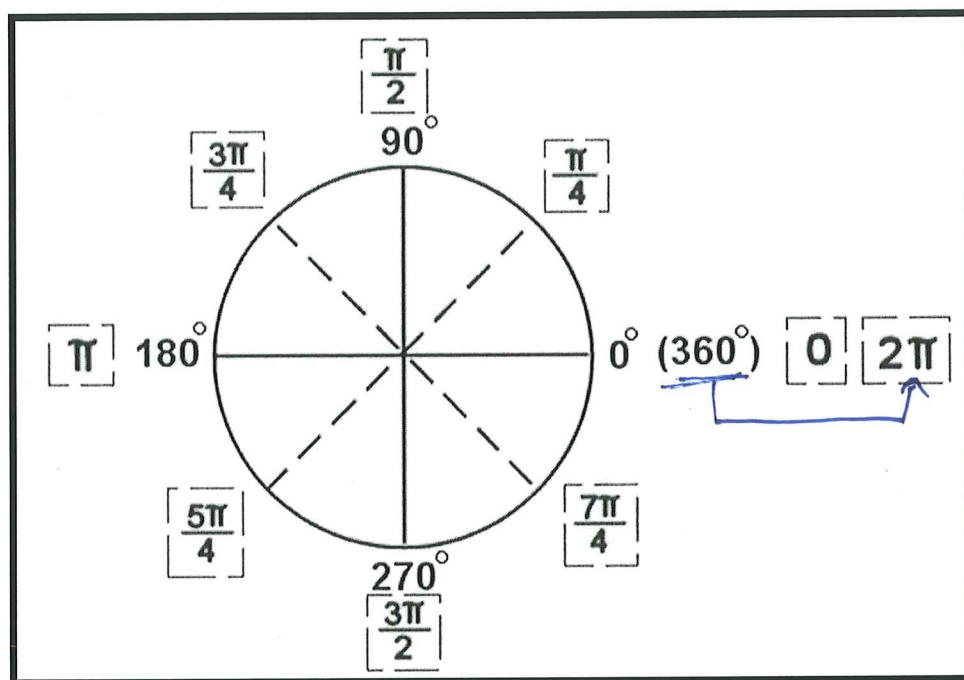
ω : $\omega = 2\pi f$ [rad/s]

$e^{j\omega t}$
↑
rad → degrees

$$e^{j\omega t} = \cos \omega t + j \sin \omega t$$



Radians & Degree



	$360^\circ = 2\pi \text{ rad}$	
	$180^\circ = \pi \text{ rad}$	60°
deg $\rightarrow$ rad:	$\varphi = \frac{\theta^\circ}{180^\circ} \cdot \pi$	
rad $\rightarrow$ deg:	$\theta = \frac{\varphi}{\pi} \cdot 180^\circ$	

$$\frac{\pi}{4} \Rightarrow \theta, \theta = \frac{\pi/4}{\pi} \cdot 180^\circ = 45^\circ.$$

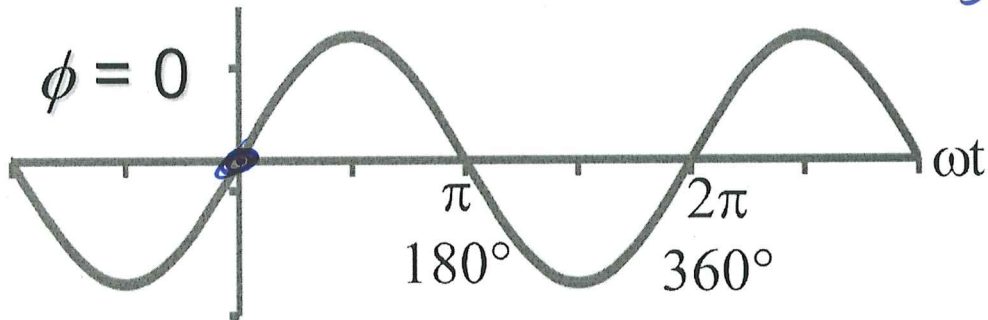
Phase angle

$$v(t) = V_m \sin(\omega t + \phi)$$

$$\sin \omega t$$

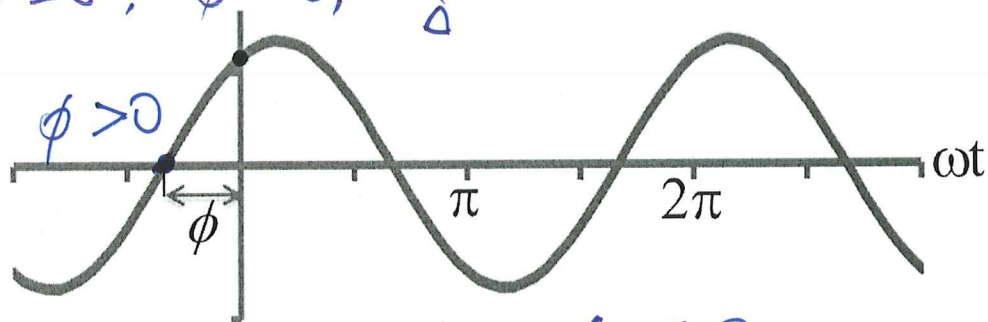
$$\sin(\omega t + \frac{\pi}{3}) \leftarrow$$

$$\sin(\omega t - \frac{\pi}{3}) \rightarrow$$



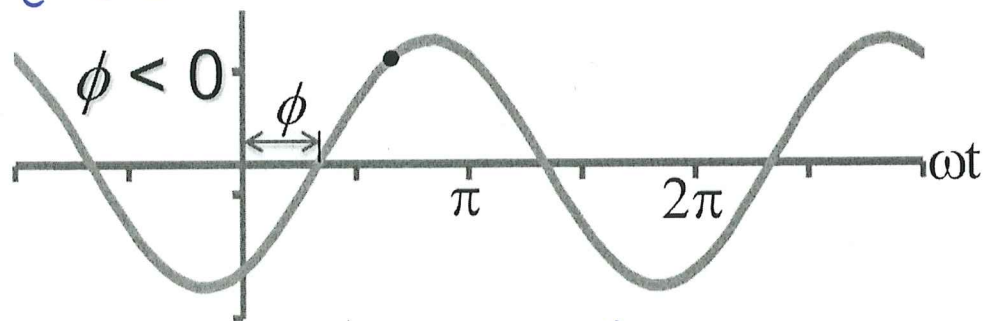
$$v(t) = V_m \sin(\omega t + \phi), \text{ when } v(t) = 0$$

$$\omega t + \phi = 0, \phi = 0, \omega t = 0. \Rightarrow t = 0.$$



$$\omega t + \phi = 0 \Rightarrow \omega t = -\phi < 0$$

$$t < 0$$



$$\omega t + \phi = 0 \Rightarrow \omega t = -\phi > 0$$

$$t > 0.$$

Trigonometric identities

$$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$$

$$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$$

$$\begin{aligned}\sin(\omega t \pm 180^\circ) &= \sin \omega t \cos 180^\circ + \cos \omega t \sin 180^\circ \\ &= -\sin \omega t\end{aligned}$$

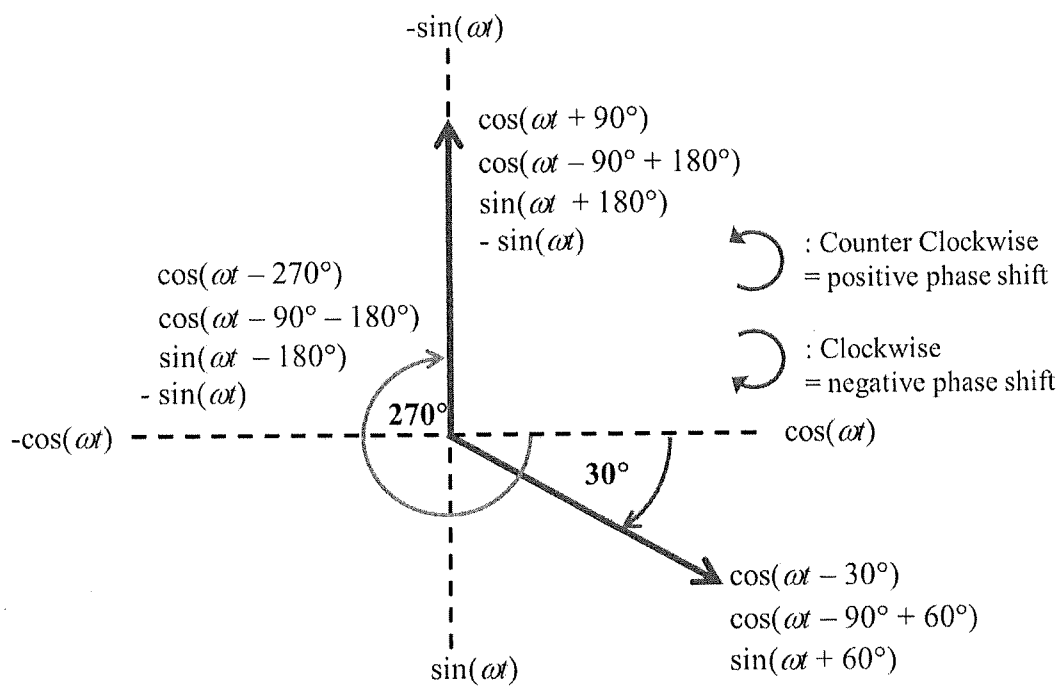
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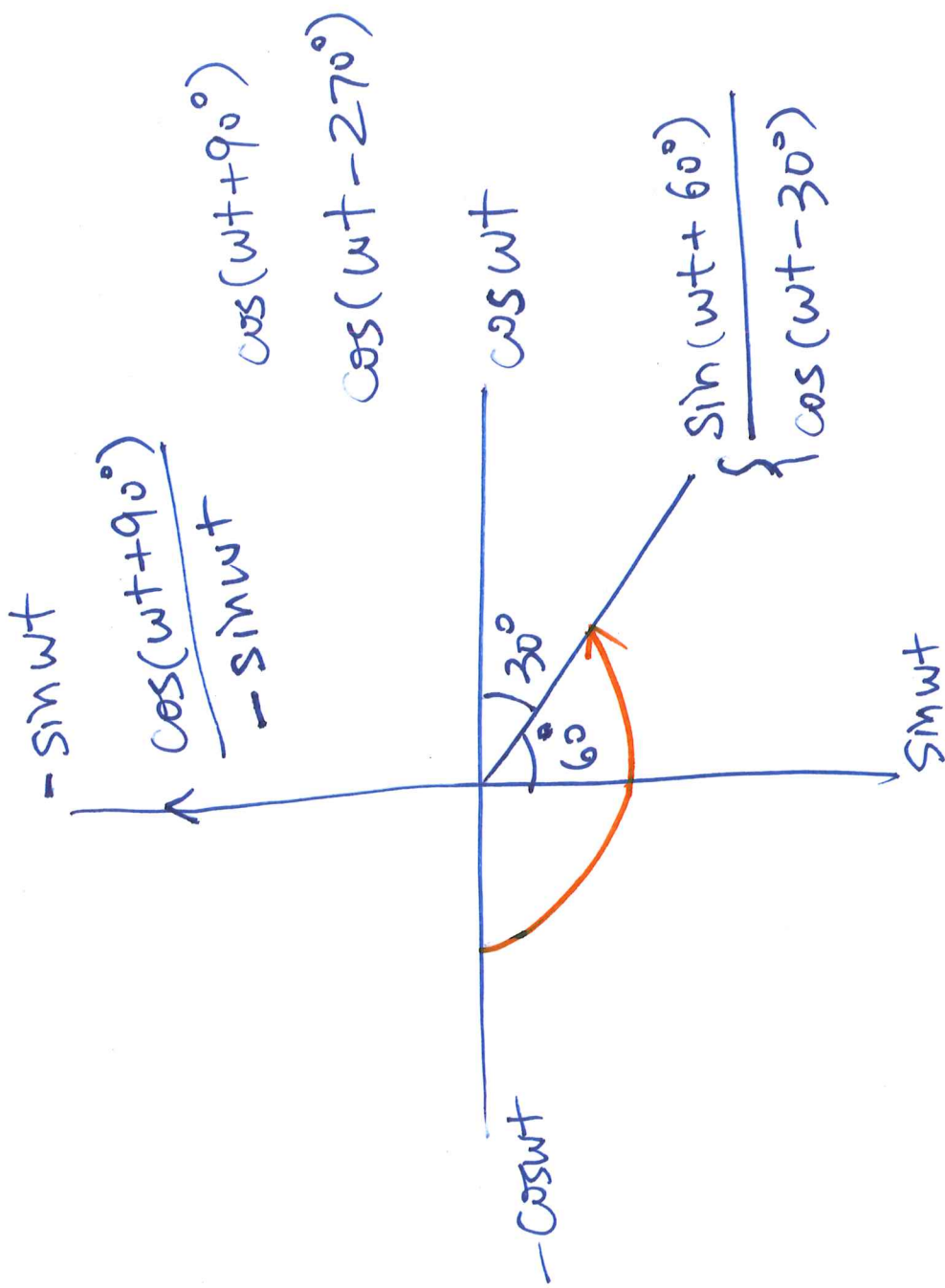
$$\begin{aligned}\cos(\omega t \pm 180^\circ) &= \cos \omega t \cos 180^\circ - \sin \omega t \sin 180^\circ \\ &= -\cos \omega t\end{aligned}$$

$$\begin{aligned}\sin(\omega t \pm 90^\circ) &= \sin \omega t \cos 90^\circ + \cos \omega t \sin 90^\circ \\ &= \cos \omega t\end{aligned}$$

$$\begin{aligned}\cos(\omega t \pm 90^\circ) &= \cos \omega t \cos 90^\circ - \sin \omega t \sin 90^\circ \\ &= -\sin \omega t\end{aligned}$$

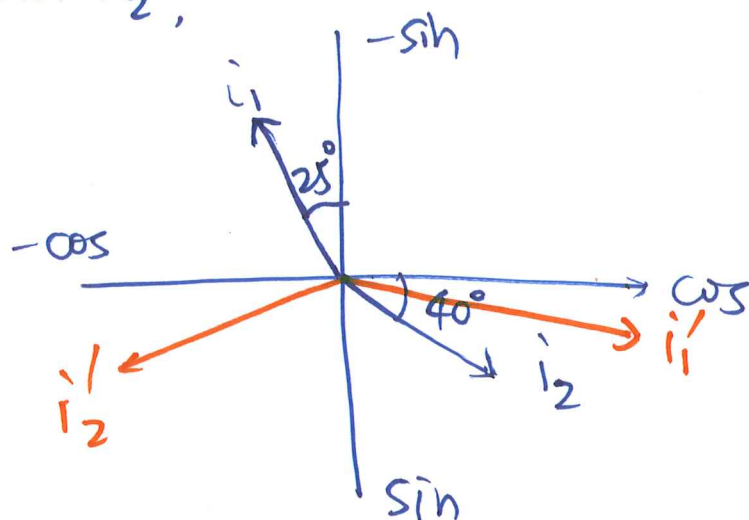
Sinusoid transform (graphic approach)





① $\sin(wt + 60^\circ)$

① transform $\rightarrow \cos$
 i_1 and i_2 ,



$$\begin{aligned} i_1 &= 4 \cos(377t + 115^\circ) \\ i_2 &= 5 \cos(377t - 40^\circ) \end{aligned} \quad \left. \begin{array}{l} 115^\circ - (-40^\circ) \\ = 155^\circ \end{array} \right\}$$

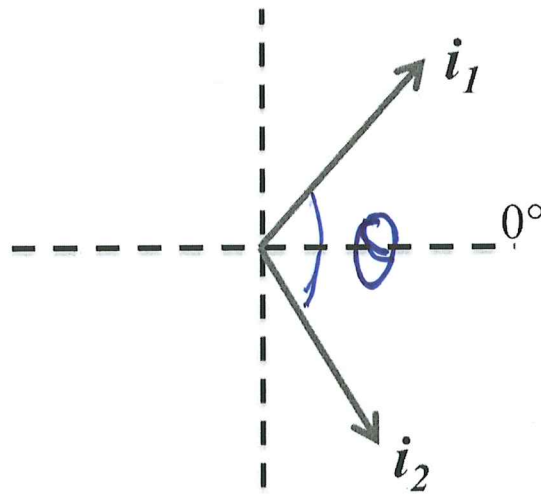
Rotate i_1, i_2 clockwise 120° .

$$i_1' = 4 \cos(377t - 5^\circ)$$

$$i_2' = 5 \cos(377t - 160^\circ)$$

i_1 leads i_2 by ~~to~~ 155° .

Sinusoid: Lead and lag



- Transform both sinusoid into cosine functions.
- Both vectors must start in bottom two quadrants.
- Rotate counter-clockwise, the one passes 0° first leads.

Example:

$$i_1 = -4\sin(377t + 25^\circ)$$

$$i_2 = 5\cos(377t - 40^\circ)$$